The first observation of the dynamical Casimir effect

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Abstract

This short seminar presents a self-contained overview of the first experimental realization of the dynamical Casimir effect, the process that consists in the conversion of particles from virtual to real by a rapidly-moving boundary condition.

1 Introduction

The static and the dynamical Casimir effects are surprising phenomena arising from the inherently quantum nature of the electromagnetic field. They both exploit the fact that "the vacuum is not empty", that is, even in the absence of any external agent the value of the electric field, while maintaining a zero mean, is constantly oscillating due to its non-zero ground energy. This phenomenon is inherently quantum-mechanical and can be seen as a manifestation of the uncertainty principle.

The Casimir effect consists in the attraction between two neutral and perfectly conductive plates. The common interpretation of this effect is that the plates impose specific boundary conditions on the field that make its zero-point energy, obtained as a renormalized sum over all modes, lower for closer plates. This effectively generates a force between the plates, that has been thoroughly verified experimentally.

In the dynamical Casimir effect, which was only theoretically predicted before 2011, one of the plates is displaced with non-uniform acceleration. This creates a mismatch in time of the vacuum modes that can generate excitations in the field. Let us discuss more precisely how this happens.

2 Particle creation from non-stationary boundary conditions

Consider a one-dimensional scalar field described by the wave equation

$$\left(\frac{1}{v^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}\right)\phi(t,x) = 0$$

which satisfies the boundary conditions

$$\begin{aligned} \phi(t,0) &= 0 \\ \phi(t,a(t)) &= 0 \end{aligned} \qquad a(t) = \begin{cases} a_0 & t \leqslant 0 \text{ and } t \geqslant T \\ \text{some } f(t) & \text{otherwise} \end{cases}.$$

The field operator in Heisenberg representation is, for $t \leqslant 0$,

$$\hat{\phi}(t,x) = \sum_n \left(\phi_n(t,x)\hat{a}_n + \phi_n^*(t,x)\hat{a}_n^\dagger\right)$$

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where the modes are

$$\phi_n(t,x) = \frac{1}{\sqrt{\pi n}} e^{-i\omega_n t} \sin\frac{\pi n x}{a_0} \qquad \text{with} \qquad \omega_n = \frac{\pi v n}{a_0}$$

[with this normalisation $H = 1/2 \int \mathrm{d}x\, (\partial_t\phi)^2 + (\partial_x\phi)^2 = \sum_n \omega_n (a_n^\dagger a_n + 1/2)$]. At any time the field operator is

$$\hat{\phi}(t,x) = \sum_n \left(\chi_n(t,x)\hat{a}_n + \chi_n^*(t,x)\hat{a}_n^\dagger\right)$$

where the functions $\chi_n(t,x)$ must satisfy both the spatial boundary conditions and the initial conditions $\chi_n(t \leqslant 0, x) = \phi_n(t,x)$. These unknown functions can be found by proposing the ansatz

$$\chi_n(t,x) = \frac{1}{\sqrt{\pi n}} \sum_m Q_{nm}(t) \sqrt{\frac{a_0}{a(t)}} \sin\frac{\pi m x}{a(t)}$$

for which the spatial boundary conditions are automatically satisfied at any moment. When this expansion is inserted into the wave equation we get a coupled system of differential equations

$$\ddot{Q}_{nm}(t) + \omega_m^2(t) Q_{nm}(t) = \sum_l \left[2\nu(t) h_{ml} \dot{Q}_{nl}(t) + \dot{\nu}(t) h_{ml} Q_{nl}(t) + \nu^2(t) \sum_l h_{lm} h_{ll} Q_{nl}(t) \right]$$

$$\omega_m(t) = \frac{\pi v m}{a(t)} \qquad \nu(t) = \frac{\dot{a}(t)}{a(t)} \qquad h_{ml} = (-1)^{m-l} \frac{2ml}{m^2 - l^2}.$$

For $t \leqslant 0$ and $t \geqslant T$ the system reduces to

$$\ddot{Q}_{nm}(t) + \omega_m^2 Q_{nm}(t) = 0.$$

For $t \leqslant 0$ the Q_{nm} are fixed by the initial conditions to $Q_{nm}(t \leqslant 0) = \delta_{nm} e^{-i\omega_n t}$. For $t \geqslant T$ there will be in general a mixing of the modes

$$Q_{nm}(t) = \alpha_{nm} e^{-i\omega_m t} + \beta_{nm} e^{i\omega_m t}$$

that changes the operators creating and annihilating each ϕ_n mode. Indeed, for $t \geqslant T$,

$$\begin{aligned}
\hat{\phi}(t,x) &= \sum_n \left(\frac{1}{\sqrt{\pi n}} \sum_m \left(\alpha_{nm} e^{-i\omega_m t} + \beta_{nm} e^{i\omega_m t}\right) \sin\frac{\pi m x}{a_0}\, \hat{a}_n + h.c. \right) \\
&= \sum_n \left(\frac{1}{\sqrt{\pi n}} \sum_m e^{-i\omega_m t} \sin\frac{\pi m x}{a_0} \left(\alpha_{nm}\hat{a}_n + \beta_{nm}^* \hat{a}_n^\dagger\right) + h.c. \right) \\
&= \sum_m \left(\frac{1}{\sqrt{\pi m}} e^{-i\omega_m t} \sin\frac{\pi m x}{a_0} \left(\sum_n \sqrt{\frac{m}{n}} \left(\alpha_{nm}\hat{a}_n + \beta_{nm}^* \hat{a}_n^\dagger\right) \right) + h.c. \right) \\
&= \sum_m \left(\phi_m(t,x)\hat{b}_m + \phi_m^*(t,x)\hat{b}_m^\dagger \right)
\end{aligned}$$

where

$$\hat{b}_m = \sum_n \sqrt{\frac{m}{n}} \left(\alpha_{nm}\hat{a}_n + \beta_{nm}^* \hat{a}_n^\dagger\right).$$

It is now clear why this process creates particles. Consider for example the case in which the field is initially in the vacuum state (for which $\hat{a}_k |0\rangle = 0$). Then the number of photons in the mth mode after the mirror has moved is

$$n_m = \langle 0 | \hat{b}_m^\dagger \hat{b}_m | 0 \rangle = \sum_n \frac{m}{n} |\beta_{nm}|^2$$

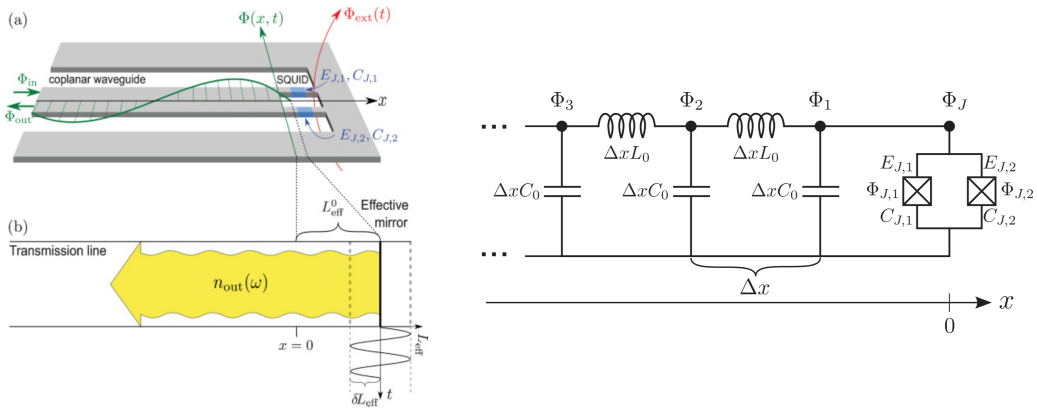


Figure 1: Schematic and distributed-element model of the experimental setup.

which in general is non-zero.

Alternatively, one could also approach the problem by analysing the behaviour of the energy-momentum tensor, in particular of the hamiltonian density, which has the advantage of being well-defined at any time. Our approach, while more intuitive, fails to give a description of the system in the non-stationary region, when there is no well-defined Fock-state basis with a time-translationally invariant vacuum state.

3 Quantum network analysis of the experimental setup

The experimental setup is composed of a superconductive coplanar waveguide terminated by a SQUID. Its schematic and the corresponding distributed element model are shown in Figure 1.

The formalism of quantum electrical networks provides a convenient way to describe the apparatus. We start by choosing a ground reference node, in our case any point on the lower branch. For every other node i we consider a branch connecting the ground node to it. For our circuit we take the branches that pass through the capacitors. Then for every node we define the (generalised) flux

$$\phi_i(t) = \int_{-\infty}^{t} \mathrm{d}t'\, v_i(t')$$

where v_i is the potential difference along the previously chosen branch connecting the ground node to the node i. In our case these are the potential differences between the ends of the capacitors. Note that this generalised flux in general is not an actual magnetic flux, since we are considering open branches. However consider a (non-generalised) magnetic flux $\phi_{\rm ext}$ passing through the SQUID, which is what is set up experimentally. This will generate an emf $\mathcal{E} = \dot{\phi}_{\rm ext}$ in the loop, for which

$$v_{J,1} - v_{J,2} = \mathcal{E} \quad \Longrightarrow \quad \dot{\phi}_{J,1} - \dot{\phi}_{J,2} = \dot{\phi}_{\rm ext} \quad \Longrightarrow \quad \phi_{J,1} - \phi_{J,2} = \phi_{\rm ext}.$$

Let us now write the constitutive equations of the electrical elements in terms of the flux between their two ends. For a capacitor the relation $q = Cv$ translates to $i = C\ddot{\phi}$, while the energy is $E = \int \mathrm{d}t\, vi = C\int \mathrm{d}t\, \dot{\phi}\ddot{\phi} = C\dot{\phi}^2/2$. For an inductor $i = \phi/L$ and $E = \phi^2/2L$. For a Josephson junction $i = I_0 \sin 2\pi\phi/\phi_0$, where $\phi_0 = h/2e$.

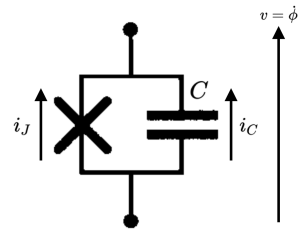
The procedure to quantize the circuit is straightforward. We start by writing a Lagrangian which is the difference between the capacitive and inductive energies of the circuit, written in terms of the fluxes. In our case, ignoring for a moment the SQUID,

$$L = \frac{1}{2}\sum_{i=1}^{\infty}\left(\Delta x C_0 \dot{\phi}_i^2 - \frac{(\phi_{i+1}-\phi_i)^2}{\Delta x L_0}\right).$$

It's easy to verify that this Lagrangian gives the expected equations of motion.

The Lagrangian for a Josephson junction is

$$L = \frac{1}{2}C\dot{\phi}^2 + E_J \cos\left(2\pi\frac{\phi}{\phi_0}\right)$$



where C is the capacitance of the junction and E_J is the Josephson energy. Again, this Lagrangian gives the expected equation of motion

$$C\ddot{\phi} = -I_0 \sin 2\pi\phi/\phi_0 \qquad \Longrightarrow \qquad -i_C = i_J = I_0 \sin 2\pi\phi/\phi_0$$

Defining $\phi_J = (\phi_{J,1} + \phi_{J,2})/2$ and $C_J = 2C_{J,1} = 2C_{J,2}$ the Lagrangian of the circuit is then

$$L = \frac{1}{2}\sum_{i=1}^{\infty}\left(\Delta x C_0 \dot{\phi}_i^2 - \frac{(\phi_{i+1}-\phi_i)^2}{\delta x L_0}\right) + \frac{1}{2}C_J\dot{\phi}_J^2 + 2E_J\cos\left(\pi\frac{\phi_{\text{ext}}}{\phi_0}\right)\cos\left(2\pi\frac{\phi_J}{\phi_0}\right)$$

from which we see that the symmetric SQUID has effectively become a single junction, with capacity C_J and tunable Josephson energy

$$E_J(\phi_{\text{ext}}) = 2E_J\cos\left(\pi\frac{\phi_{\text{ext}}}{\phi_0}\right).$$

Legendre-transforming the Lagrangian into the Hamiltonian allows us to write the Heisenberg equation of motion for the ϕ operator that is, in the $\Delta x \to 0$ limit,

$$\left(\frac{1}{v^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}\right)\phi = 0$$

where $v = 1/\sqrt{C_0 L_0}$. The equation of motion for the conjugate momentum becomes a boundary condition for the flux at the end of the transmission line

$$C_J\frac{\partial^2\phi}{\partial t^2} + \left(\frac{2\pi}{\phi_0}\right)^2 E_J(t)\phi(0,t) + \frac{1}{L_0}\frac{\partial\phi}{\partial x}(0,t) = 0.$$

If the external magnetic flux through the SQUID is constant, $E_J(t) = E_J$, the boundary condition can be rewritten as

$$\omega_s^2\frac{\partial^2\phi}{\partial t^2} + \phi(0,t) + L_{\text{eff}}\frac{\partial\phi}{\partial x} = 0$$

where $\omega_s^2 = (2\pi)^2 E_J/\phi_0^2 C_J$ is the so-called plasma frequency of the SQUID and $L_{\text{eff}} = \phi_0^2/(2\pi)^2 E_J L_0$. In the regime $\omega \ll \omega_s$

$$\phi(0,t) + L_{\text{eff}} \frac{\partial \phi}{\partial x} = 0,$$

that becomes, for $k \ll 1/L_{\text{eff}}$ or equivalently $\omega \ll v/L_{\text{eff}}$, the mirror-like boundary condition

$$\phi(L_{\text{eff}}, t) = 0.$$

Therefore the experimental setup effectively generates a boundary condition for the field. Moreover, we expect this boundary condition to be variable for a non-static ϕ_{ext}.

4 Properties of the emitted radiation

In the same spirit of the first section we can quantize the modes entering and exiting the transmission line

$$\phi_\omega(t,x) = e^{-i(\pm kx + \omega t)},$$

now indexed by the continuous frequency ω. This is due the fact that we are imposing only one boundary condition.

If the flux is varied harmonically at a frequency ω_d, one finds that, for $\omega < \omega_d$,

$$\hat{b}(\omega) = R(\omega)\hat{a}(\omega) + S(\omega)\hat{a}^\dagger(\omega)$$

$$R(\omega) = \exp\left(-2ikL_{\text{eff}}(\phi_{\text{ext}})\right) \qquad S(\omega) = -i\frac{\delta L_{\text{eff}}}{v}\sqrt{\omega\left(\omega_d - \omega\right)}A(\omega)A^*\left(\omega_d - \omega\right).$$

$R(\omega)$ is the expected reflection upon the SQUID effective mirror, while $S(\omega)$, mixing the $\hat{b}$ and $\hat{a}^\dagger$ operators, is responsible for the DCE. With this expression we can calculate the output photon flux density for an input thermal state

$$n_{\text{out}}(\omega) = \langle \hat{b}^\dagger(\omega)\hat{b}(\omega)\rangle = n_{\text{in}}(\omega) + |S(\omega)|^2 n_{\text{in}}(\omega_d - \omega) + |S(\omega)|^2$$

The first two terms, proportional to the thermal occupation number $n_{\text{in}}(\omega)$, represent the purely classical effects of reflection and upconversion of the input field to the drive frequency. They vanish at zero temperature. The last term on the right-hand side is due to vacuum fluctuations and is, in fact, the DCE radiation. From this expression we also see that the spectrum is symmetric around the drive frequency ω_d and quite broad. These two features are verified experimentally.

Another peculiar characteristic of the emitted radiation resides its correlation properties. Heuristically, the non-vanishing expectation value

$$\left\langle b\left(\frac{\omega_d}{2} - \Delta\omega\right) b\left(\frac{\omega_d}{2} + \Delta\omega\right)\right\rangle \neq 0,$$

while not directly measurable, can be interpreted as the correlation between two photons that are simultaneously created at frequencies $\omega_d/2 \pm \Delta\omega$. This is corroborated by the fact that the second-order coherence function $g_2(\tau)$ is always greater than one, and $g_2(0) \gg 1$. This large value at $\tau = 0$ can be seen as a consequence of the production in pairs of the photons, so that the probability of detecting two photons simultaneously is basically the same as the probability of detecting one. Experimentally this correlation cannot be measured by a simple coincidence-counter, since single-photon counters working at the microwave frequencies are not available.

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